Beta regression for the microbiome data

Chapter 4.3.3: Generalized linear models

This model regresses a measure of microbiome species richness onto features of the home from which the sample was taken. Let OTU_{ij} be the abundance of Operational Taxonomic Unit (OTU) j in sample i. The response variable is the proportion of the abundance attributed to the most abundant OTU,

$$Y_i = \frac{\max\{OTU_{i1}, \ldots, OTU_{im}\}}{\sum_j OTU_{ij}} \in (0, 1).$$

There are eight covariates (X_{ij}):

1. Longitude
2. Latitude
3. Annual average temperature
4. Annual average precipitation
5. Net primary production
6. Elevation
7. The binary indicator of whether it is a single-family home
8. Number of bedrooms

The regression model is

$$Y_i \sim \text{Beta}(rq_i, r(1-q_i)) \text{ where } \text{logit}(q_i) = \sum_{l=1}^{p} X_{ij}\beta_j,$$

so that the expected value of Y_i is $q_i \in [0, 1]$ and the concentration around q_i is determined by $r > 0$. The regression coefficients have uninformative priors $\beta_j \sim \text{Normal}(0, 10^2)$ and the concentration parameter has prior $r \sim \text{Gamma}(0.1, 0.1)$.

Load the data

```
set.seed(0820)

load("S:\\Documents\\My Papers\\BayesBook\\Data\\Microbiome\\homes.RData")
ls()
```

```
## [1] "homes" "OTU"
```

```
city     <- homes[,2]
state    <- homes[,3]
lat      <- homes[,4]
long     <- homes[,5]
temp     <- homes[,6]
precip   <- homes[,7]
NPP      <- homes[,8]
elev     <- homes[,9]
house    <- ifelse(homes[,10]=="One-family house detached from any other h
bedrooms <- as.numeric(homes[,11])
```

```
## Warning: NAs introduced by coercion
```

```
OTU      <- as.matrix(OTU)
Y        <- apply(OTU,1,max)/rowSums(OTU)
X        <- cbind(long,lat,temp,precip,NPP,elev,house,bedrooms)
names    <- c("Intercept","Longitude","Latitude",
              "Temperature","Precipitation","NPP",
              "Elevation","Single-family home",
              "Number of bedrooms")

# Remove observations with missing values
junk        <- is.na(rowSums(X))
Y           <- Y[!junk]
X           <- X[!junk,]
city        <- city[!junk]
state       <- state[!junk]

# Standardize the covariates
X           <- as.matrix(scale(X))

X           <- cbind(1,X) # add the intercept
colnames(X) <- names
n           <- length(Y)
p           <- ncol(X)
```

Plot the data

```
hist(Y,breaks=50,ylab="Maximum proportion")
```

Fit the beta regression model in JAGS

```
library(rjags)

data   <- list(Y=Y,X=X,n=n,p=p)
params <- c("beta","r")

model_string <- textConnection("model{
   for(i in 1:n){
     Y[i]       ~ dbeta(r*mu[i],r*(1-mu[i]))
     logit(mu[i]) <- inprod(X[i,],beta[])
   }
   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
   r ~ dgamma(0.1,0.1)
}")

model <- jags.model(model_string,data = data, n.chains=2,quiet=TRUE)
update(model, 10000, progress.bar="none")
samples <- coda.samples(model, variable.names=params, thin=5, n.iter=20000

plot(samples)
```